MIL-T-81521A(AS)
24 April 1969
SUPERSEDING
MIL-T-81521(AS)
2 March 1967

MILITARY SPECIFICATION

TAPE, TEXTILE, NYLON, LANDBASED ARRESTING GEAR, 7 INCH

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification covers the requirements of a 7 inch nylon tape having a breaking strength of not less than 70,000 pounds.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

CCC-T-191	Textile Test Methods

Military

MIL-B-13239	Barrier Material, Waterproofed, Flexible, All Temperature
MIL-B-81374	Barrier Material, Greaseproof, Waterproof, Flexible, High Strength

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

FSC 8315

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DRAWINGS

Naval Air Engineering Center

B506730	Tape Reel Assembly, Tape Installation Method (7 Inch Tape)
508306	Reel, Tape Assembly

(When requesting specifications, standards, and publications, refer to both title and number. Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. All other documents should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Rules and Regulations under the Textile Fiber Products Identification Act.

(Application for copies should be addressed to the Federal Trade Commission, Washington, D.C. 25380.)

U.S. Air Force Specification Bulletin No. 90 Color Codes Manufacturers' Identification, Nylon Webbing.

(Application for copies should be addressed to the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials -

* 3.1.1 Yarn - The yarn used in the manufacture of the tape shall be 840 denier, 8.5 grams per denier minimum, continuous 140 filament, bright, high-tenacity, light and heat resistant nylon. It shall be a polyamide prepared from hexamethylene diamine and adipic acid, or its derivatives, and shall have a minimum melting point of 244° C. Yarn designed for cordage use is not permitted in ground and binder warps. The yarn shall be substantially free from sizing, loading, or other adulterants, and shall not be bleached in any manner or by any process.

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3.2 Weave - The weave shall be as shown in Figure 1.

3.3 Color - The color of the tape shall be black as produced by the finishing treatment.

3.4 Construction and physical properties - The finished nylon tape, after resin treatment, shall conform to the construction and physical properties of Table I.

TABLE I

PHYSICAL PROPERTIES

CHARACTERISTIC	REQUIREMENT
Width, inches	7.0 ±0.062
Thickness, inches	0.192 ±0.005
Weight, ounces/yard (max)	24.5
Breaking Strength, pounds (min)	70,000
Ends in warp (min)	
Ground	349
Binder	78
Stuffer	871
Filling, yarns per inch (min)	19
Yarn ply (min)	
Ground	3
Binder	2
Stuffer	5
Filling	5

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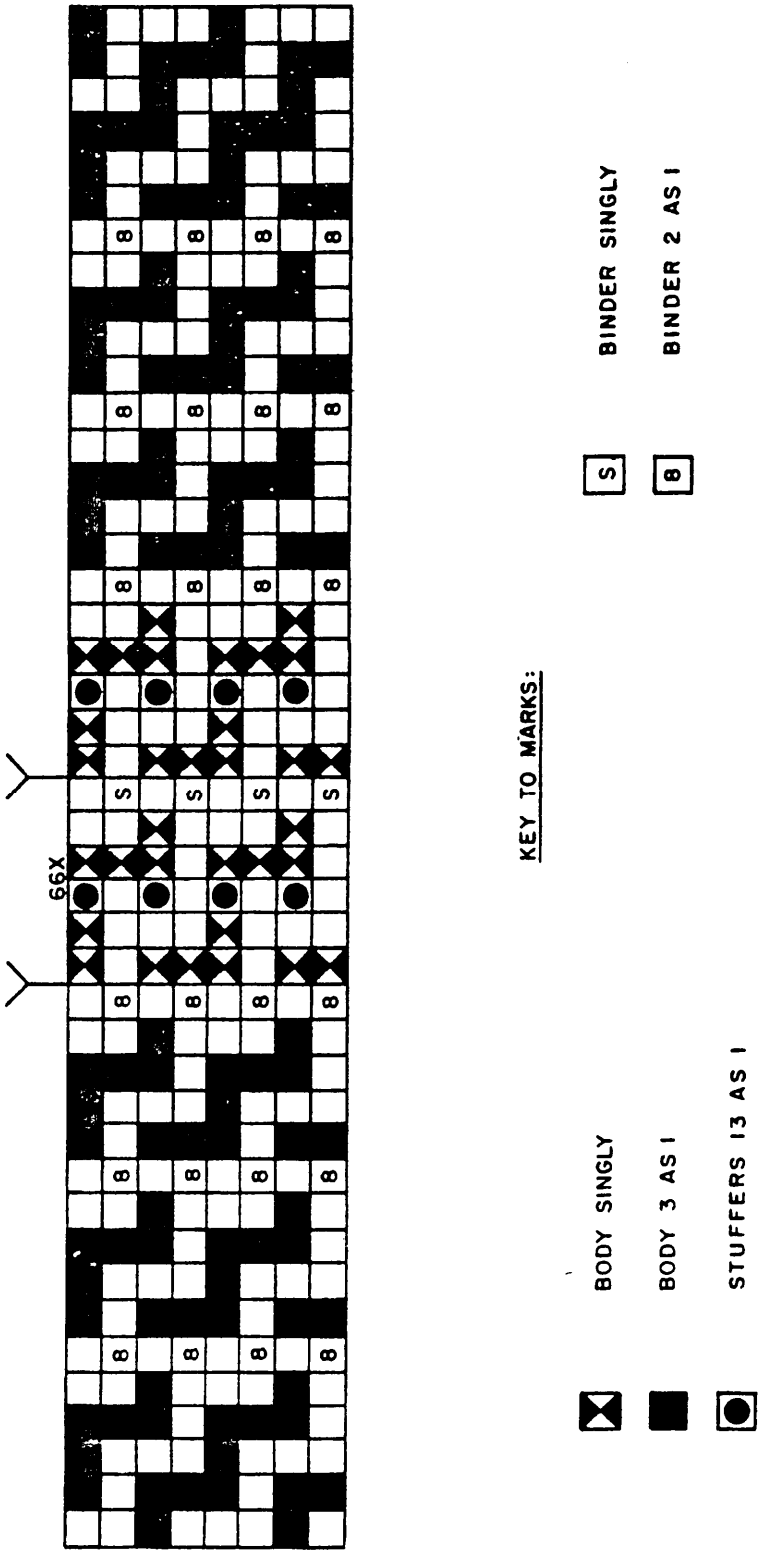


Figure 1. Tape Weaving Pattern

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TABLE I (Continued)

CHARACTERISTIC	REQUIREMENT
Ply twist, turns per inch	
Ground	2.0 - 3.0
Binder	2.0 - 3.0
Stuffer	1.0 - 2.0
Filling	1.0 - 2.0

3.5 Length of roll - Unless otherwise specified, each roll of tape shall contain either 1375 plus 0 minus 5 foot or 1500 plus 0 minus 5 foot lengths as specified in the contract or purchase order. No roll shall contain more than one piece. (See 4.5.2.3 for one additional 25 foot sample at start of each loom run.) (See 5.2 for packing of referenced two lengths.)

3.6 Resin treatment of tape -

3.6.1 The resin shall be polyvinyl butyral plasticized with butyl ricinoleate suitably applied from a water dispersion, dried, and cured so as to form a firmly adherent and evenly distributed deposit or coating on the yarns of the tape. The curing of the resin impregnated tape shall be done in the temperature range of 240 to 360° F (115 to 182.4° C).

3.6.2 The tape shall contain between 4 to 8.5 percent by weight of matter extractable in methyl-ethyl-ketone.

3.6.3 Fine carbon black shall be added to the resin to produce a uniform black color.

3.7 Identification -

3.7.1 Manufacturer's identification - Each manufacturer shall incorporate as part of the binder warp at the center of the tape two ends dyed to match the shade assigned to the manufacturer in accordance with Air Force Bulletin No. 90. A producer not listed in Bulletin No. 90 shall apply to the procuring activity for assignment of color coding. The colorfastness of this yarn shall be rated as "Fair" or "Good" when tested by method as specified in Table II.

3.8 Fiber identification - Each roll of tape shall be labeled, ticketed, or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act.

3.9 Workmanship - The finished tape shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the acceptable levels specified herein.

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4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Inspection of components - Quality conformance provisions for components described shall be in accordance with subsidiary specifications and drawings referenced to the extent applicable, except that this specification shall govern in the event of conflict. In addition, the supplier shall furnish a certificate of compliance for the characteristics referenced in 3.1.1. The certificate of compliance shall be accompanied by actual test, inspection or other verifiable quality data.

* 4.3 Tape shall be tested and inspected as noted in 4.4 and must pass the requirements shown in Table I.

* 4.4 Quality conformance inspection - The quality conformance inspection shall consist of:

(a) Sampling Plan A and tests.

(b) Sampling Plan B and tests.

(c) Sampling Plan C and tests.

(d) Sampling Plan D and tests.

(Tests for each Sampling Plan are those shown in Table II alongside the appropriate plan letter with the test methods as shown therein.)

* 4.4.1 Sampling Plan A - Tests are to be run on every 1375 or 1500 foot reel of tape.

* 4.4.2 Sampling Plan B - Tests shall be made on one sample for each continuous loom run.

* 4.4.3 Sampling Plan C - Tests shall be run on samples taken prior to weaving from every 10,000 pounds of yarn or fraction thereof.

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4.4.4 Sampling Plan D - For this inspection of product, sampling shall be conducted in accordance with 4.4.4.1.

4.4.4.1 Inspection levels - The inspection levels shall be as follows:

(a) Yard-by-yard examination - A one yard unit of tape shall be inspected at intervals of approximately 40 feet as indicated in 4.5.4.1 and 4.5.4.1.1 prior to addition of resin as specified in 3.6.

(b) Overall examination - The pieces examined shall be those selected for yard-by-yard examination.

4.5 Test methods -

4.5.1 The methods of testing specified in CCC-T-191, wherever applicable, and as listed in Table II, and the following sub-paragraphs shall be followed. The physical and chemical values specified in Section 3 apply to the average of the determinations made on a unit of product for test purposes as specified in the applicable test methods. Material not conforming to the test requirements shall reject the lot represented.

TABLE II

TESTS AND TEST METHODS

SAMPLING PLAN	TEST	METHOD	
		CCC-T-191 METHOD NO.	OTHER
A	Width	5020	
A	Thickness	5030 1/	
A	Weight	5040	
A	Length	5010	
A	Breaking strength		4.5.2
A	Extractable matter		4.5.3

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TABLE II (Continued)

SAMPLING PLAN	TEST	METHOD	
		CCC-T-191 METHOD NO.	OTHER
B	Ends in warp		
	Ground		Visual
	Binder		Visual
	Stuffer		Visual
B	Filling, yarns per in.	5050	
B	Yarn ply		
	Ground		Visual
	Binder		Visual
	Stuffer		Visual
	Filling		Visual
B	Twist, turns per in.		
	Ground	4054.1 4/	
	Binder	4054.1 4/	
	Stuffer	4054.1 4/	
	Filling	4054.1 4/	
B	Weave		Figure 1
C	Colorfastness to laundering	4614 2/, 3/	
C	Melting point	1534 5/	
C	Yarn strength	4100 5/	

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TABLE II (Continued)

SAMPLING PLAN	TEST	METHOD	
		CCC-T-191 METHOD NO.	OTHER
C	Denier	4021 5/	4.6
C	Filament count	5/	Visual, 4.6
D	Classification of defects		4.5.4
	Yard-by-yard		4.5.4.1
	Overall		4.5.4.2

1/ Except that a six ounce total load shall be applied and the presser foot diameter shall be 3/8 inch.
2/ Applicable only to manufacturer's identification yarns.
3/ Except that nylon color transfer cloth shall be used.
4/ One test on each 5 yarns.
5/ Unless otherwise specified, a supplier's certificate of compliance is required and will be accepted for the requirement.

4.5.2 Breaking strength - The breaking strength test shall be conducted in accordance with 4.5.2.2. The type and capacity of the tensile testing machine shall be suitable for the type of tape undergoing tests. The machine shall be adjusted so that no-load separation of the grips or pulling jaws shall be at the rate of travel specified in 4.5.2.1. All breaking tests shall be made on full width sections of tape.

4.5.2.1 The test grips (see 6.3) for holding the specimen shall be of the split-drum type approximately 6-1/2 inches in diameter and 10-1/2 inches in length or approved equal. The no-load rate of jaw separation shall be 4 - 8 inches per minute. The distance between the centers of the holding drum at the start of the test shall be 3 feet 6 inches.

4.5.2.2 Ultimate strength test - A test specimen 25 feet long shall be cut from each tape as noted in 4.5.2.3. Three samples 7 feet 6 inches long cut from the 25 foot section shall be load tested. A failure above 70,000 pounds tension or a jaw break above 70,000 pounds will be considered acceptable. A

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failure below 70,000 pounds tension or a jaw break below 70,000 pounds will be considered unacceptable. Two satisfactory sample tests of the three samples tested shall be considered proof of satisfactory tape strength of the specimen.

4.5.2.3 Method of sampling for breaking strength test - Contract or order will specify either 1375 or 1500 foot lengths, or both. The actual delivered length shall be 1350 or 1475 feet after removal of a 25 foot sample from the tapes. The first section of tape leaving a loom run shall have a 25 foot sample removed from the lead end. After this every 1375 foot or 1500 foot section shall have 25 feet removed from the end on the outside of the roll of tape. The rolls of tape must be marked consecutively as they come off the loom with the 25 foot samples marked identically with the rolls of tape to which they apply. The tape samples shall be tested as described in 4.5.2.2. Failure of a sample to pass the test in 4.5.2.2 will cause rejection of the 1375 foot or 1500 foot roll of tape following the sample. (Any deviation from the described procedure may only be made upon approval by the Commanding Officer, Naval Air Engineering Center, Attn: NE-7, Philadelphia, Pennsylvania 19112.)

4.5.3 Extractable matter - The extractable matter, inclusive of the resin deposit, shall be determined on a specimen of approximately 10 grams of the tape. The sample, after separation of the warp and filling yarns, shall be dried to constant weight in a weighing bottle at a temperature of 220 ±8°F (104 ±4.5°C). After six hour extraction with methyl ethyl ketone in Soxhlet apparatus, the final weight of the extracted sample shall be obtained after constant weight has been obtained under the same drying conditions as before.

$$\text{Percent extractable matter} = \frac{\text{Loss in weight on extraction}}{\text{Original dry weight of sample}} \times 100$$

4.5.4 Classification of defects - Defects found during the examination shall be classified in accordance with 4.5.4.1 through 4.5.4.2. The Government reserves the right to require examination for any defects not included herein and to classify such defects in accordance with the definitions contained in MIL-STD-105.

4.5.4.1 Yard-by-yard - The required yardage of each piece shall be examined and visual defects classified as listed in Table III. The defects found shall be counted regardless of their proximity to each other except where two or more defects represent a single local condition of the tape, in which case only the more serious defects shall be counted. A continuous defect shall be counted as one defect for each warpwise yard, or fraction thereof, in which it occurs. The unit of product for this examination shall be one linear yard.

4.5.4.1.1 Yard-by-yard acceptance quality level - For critical defects the acceptance number shall be zero defects. For major defects the AQL shall be 2.5 defects per 100 units and 10.0 total defects per 100 units (i.e. yards).

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TABLE III

CLASSIFICATION OF DEFECTS

DEFECTS	DESCRIPTION	CRITICAL	MAJOR	MINOR
Abrasion Marks	Resulting in rupture of yarns, or in nap sufficient to obscure the identity of any yarn exceeding 10 percent of width or one inch in length.	X		
Broken or Missing End	Two or more regardless of length of a single end exceeding 6 inches in length.	X		
	Single end under 6 inches but exceeding 1/4 inch.			X
Broken or Missing Pick	Two or more regardless of extent. NOTE: The filling tie-in or jointing shall not be construed as a defect of any nature.	X		
Coarse or Light Filling Bar	Resulting in visible difference in stiffness or thickness of tape and extending for more than 1/4 inch or less in the length direction.	X		
Twist or Distortion	Tape will not lay flat upon application of manual pressure due to twist or distortion.			X
Cut, Hole or Tear	Any cut, hole or tear.	X		
Drop-Ply	Clearly visible on more than two ends within same length and extending over 9 linear inches or more.		X	

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TABLE III (Continued)

CLASSIFICATION OF DEFECTS

DEFECTS	DESCRIPTION	CRITICAL	MAJOR	MINOR
Edge Loopy	Forming clearly visible filling loops, or edges tied loosely to body of webbing for 2 linear inches or more.	X		
Edge Cut, Torn or Frayed	Clearly visible along edge.	X		
Edge Scalloped	Any indentation of edges above allowable tolerances.		X	
Floats or Skips	Multiple, 1/2 inch or more in combined warp and filling direction or single float or skip over more than 1 inch. Multiple, less than 1/2 inch in combined warp or filling directions or single float or skip over more than 1/2 inch, but not exceeding 1 inch if in warp, or more than 1/4 inch of the width but not exceeding 1 inch, if in filling.			X
Hitchback, Crack, Open-place	Clearly visible opening between adjoining picks, or warpwise tension area over part of the width resulting in visible light and heavy place.	X		
Jerked-In, Filling Slough-Off, Slug	A clearly visible loop of filling pulled in at edges.			X
Kinks	More than 3 in any 9 linear inches.		X	
Knots	More than 1 knot in any 5 linear inches. One knot every 2 yards with untrimmed ends extending from surface of tape.		X	

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TABLE III (Continued)

CLASSIFICATION OF DEFECTS

DEFECTS	DESCRIPTION	CRITICAL	MAJOR	MINOR
Mispick, Double Pick	Two or more across the full width. Single across the full width.			X
Slack End	Two or more in the same length, jerked in between picks, or forming clearly visible loops on the surface.		X	
Slub or Slag, Gout	More than twice the thickness of the yarn (or ply, if plied).			X
Smash	Any smash	X		
Spot, Stain or Streak	Any clearly visible dirt, rust, grease, oil, spot, stain or streak.			X
Wrong Draw	Extending for more than 9 inches.		X	
Identification Threads	Misplaced. Missing.	X X		

4.5.4.2 Overall - Examine required samples for finish, odor and cleanliness. The finish must be equal to the approved sample (when available) (see 6.4). There must not be any objectionable odor and the samples must be clean.

4.5.4.3 Examination of preparation for delivery requirements - An examination shall be made to determine that packaging, packing and marking requirements of Section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The Inspection Level shall be S-2 and AQL shall be 4.0 defects per 100 units.

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EXAMINE	DEFECT
Marking	Omitted, illegible, location, sequence, or method of application.
Materials	Any component missing. Any component damaged, affecting serviceability.
Workmanship	Inadequate application of components. Bulging or distortion of containers.

4.6 Denier and filament - An affidavit from the yarn manufacturer will be acceptable as proof that denier and filament requirement are met.

5. PREPARATION FOR DELIVERY

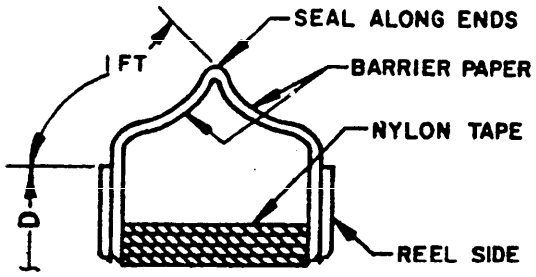
5.1 Packaging - Not applicable.

* 5.2 Packing - The tape shall be installed on a reel as shown on NAEC drawing B506730. The reel shall be manufactured as shown on NAEC drawing 508306, identified as Part Number 508306-3 for lengths specified in 3.5. When installing tape on reel, it shall be installed over barrier material conforming to MIL-B-81374 or MIL-B-13239, Type B-2, Class 1, Grade A. The barrier material shall also cover the sides of the tape reel (inside of the shipping reel) with two additional feet of paper in excess of the reel side diameters to permit sealing the two sides together all around the reel (see Figure 2). The additional barrier material shall permit the opening and re-sealing of the barrier paper if examination of or work on the tape is required. Sealing process for barrier material shall not injure the tape. After installing the tape on reel and covering with barrier material, wooden lagging shall be installed around edge of reel, as shown on NAEC drawing B506730.

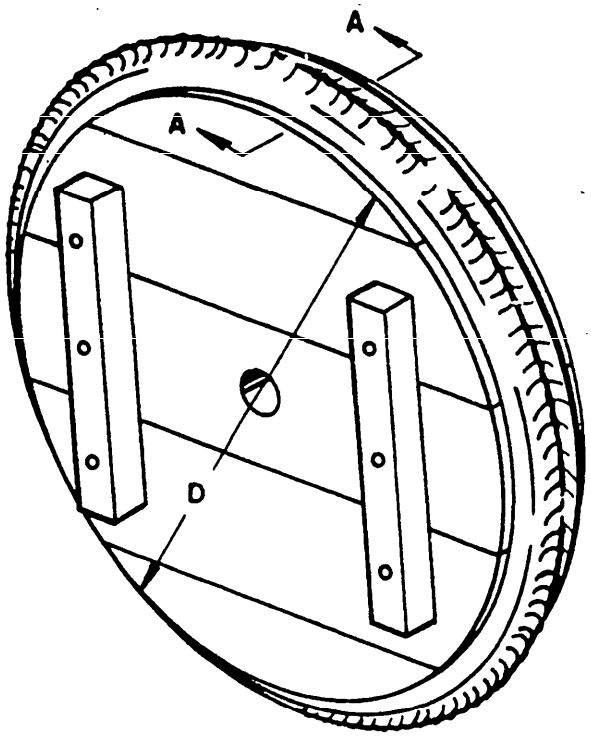
5.3 Marking - Each reel shall be durably and legibly marked as follows:

TAPE, TEXTILE, NYLON, LANDBASED ARRESTING GEAR
Specification MIL-T-81521A(AS)
Width
Yardage
Manufacturer's Identification Number for the material
Contract or Order No.
Stock No.
Date of Manufacture
Name of Supplier
Name of Manufacturer (if not the same as supplier)

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SECTION A-A



D = DIA OF REEL SIDE

Figure 2

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In addition, each reel shall be marked for shipment in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use - The nylon tape covered by this specification is intended for use in landbased aircraft arresting gear.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Quantity desired.

(c) Manufacturer's identification.

(d) Color required if other than specified in 3.3.

(e) Length of roll required (1375 feet or 1500 feet).

6.3 Specifications and drawings of the test grips required in testing are available on application to Commanding Officer, Naval Air Engineering Center, Attn: NE-7, Philadelphia, Pennsylvania 19112 for drawing 505102.

6.4 Marginal notations - The margins of this specification are marked with an asterisk to indicate where changes (deletions, additions, etc.) from the previous issue were made. This was done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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DOD Project No. 8315-N140

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-T-81521A(AS) TAPE, TEXTILE, NYLON, LANDBASED ARRESTING GEAR, 7 INCH

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED *(Printed or typed name and activity)*	DATE

DD FORM 1 NOV 64 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

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Naval Air Engineering Center
Philadelphia, Pennsylvania 19112

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